

EMMA VEGA

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EDUCATION

University of California, Los Angeles

Expected June 2027

B.S. in Mathematics/Applied Science — Physics concentration

GPA **3.41**

RESEARCH EXPERIENCE

Percolation in Soft Random Geometric Graphs

April 2025 - Present

UCLA Mathematics

University of California, Los Angeles

- Supervisors: Professor Mason Porter and James Von Raj (physics graduate student)
- Conducted literature review in network theory and percolation theory focusing on configuration models, percolation, and network resistance
- Implemented numerical simulations of Erdős–Rényi, Barabási-Albert, and configuration model with power law degree sequence graphs in Python using NetworkX
- Modeled the largest connected component as a function of node occupancy and mean degree with Matplotlib
- Future: investigate differences in percolation thresholds between random geometric graphs and Erdős–Rényi graphs

A Scalable Workflow for Curating and Classifying Diffraction Images

June 2026 - August 2026

CICF Summer Program

Cornell High Energy Synchrotron Source (CHESS) at Cornell University

- Supervisors: Dr. Suchi Sarker and Dr. Valentin Kuznetsov
- Will present research at CICF Virtual Summer Symposium

National Science Foundation Cyberinfrastructure Compass Fellowship (NSF CICF)

January - April 2026

Student Fellow

NSF Cyberinfrastructure Center of Excellence

- Virtual training program about cyberinfrastructure, i.e. the computing systems, data storage systems and repositories, advanced instruments, and visualization environments, to improve research productivity and enable scientific breakthroughs
- Developed technical skills: command line, SciPy, scientific computing, Git, cloud computing, data workflows, classical machine learning
- Presented with 5 other fellows to 30 people a group research project about an NSF Major Facility, Vera C. Rubin Observatory, and its scientific mission, cyberinfrastructure, and data lifecycle

Calculating the Hubble Constant with “Strong” Gravitational Lensing

July - August 2022

California State Summer School for Mathematics and Science (COSMOS)

University of California, Davis

- Supervisor: Professor Chris Fassnacht
- Analyzed “strong” gravitational lensing data using Python (NumPy, Matplotlib) in Jupyter Notebook
- Identified trends relevant to calculating the Hubble constant
- Presented results via Powerpoint oral presentation and poster board

WORK EXPERIENCE

California State Summer School for Mathematics and Science (COSMOS)

July - August 2024

Residential Assistant

University of California, Los Angeles

- Served as co-Residential Assistant for 20 rising junior and senior high school students, mentoring students interested in physics and aerospace or mechanical engineering

- Planned and led nightly recreational activities and supervised weekend field trips
- Supported student welfare, including conflict mediation, crisis management, and overnight on-call responsibilities

VOLUNTEER EXPERIENCE

Southern California FIRST Tech Challenge Robotics

Referee, Robot Inspector, Queuer, Mentor

November 2023 - Present

Greater Los Angeles Metropolitan Area, CA

- Collaborated with a team of 5 referees to make real-time gameplay decisions in a fast-paced environment
- Inspected robots for mechanical compliance with competition rules
- Troubleshoot software used for robot inspection and scorekeeping
- Coordinated 3 volunteers to ensure 42 teams were ready for matches
- Interviewed by FTC Team 4153 Dead Space to discuss experience as a FIRST alum

WORKSHOPS AND CONFERENCES

FUTURE of Physics @ Caltech

California Institute of Technology

September 2025

Pasadena, CA

- Program for promising rising junior or senior undergraduates offering an in-depth look into physics graduate school

Leadership Alliance Virtual Professional Development Series

Brown University

June to July 2025

Virtual

Computational Research Access NEtwork (CRANE) Seminar

National Science Foundation

January to May 2024, April 2025

Virtual

- Developed computational research skills in plasma physics and sciences using Python
- Created basic physics simulations from scratch using numerical integration and finite difference methods
- Applied introductory machine learning, signal and image processing; introduced to Windows Terminal and Git

APS Conference for Undergraduate Women and Gender Minorities in Physics (APS CU*IP)

California Polytechnic State University

January 2025

San Luis Obispo, CA

Computing Across the Sciences Mini Semester

Argonne National Laboratory

November 2024

Virtual

Workshop in Plasma Physics for Undergraduates

Princeton Plasma Physics Laboratory (PPPL)

June 2024

Princeton, NJ

- Workshop for rising sophomore and junior students about plasma physics, fusion, and the research done at PPPL
- Interviewed for article covering the program

American Physical Society Conference for Undergraduate Women in Physics (APS CUWIP)

University of San Diego

January 2024

San Diego, CA

ACTIVITIES

UCLA Women in Math

Undergraduate Mentee

October 2025 - Present

University of California, Los Angeles

LGBTQ Resource Center QTBIPOC Space

Member

February 2024 - Present

University of California, Los Angeles

UCLA Women+ in Physical Sciences

Member, Community Town Hall Speaker & Discussion Leader

November 2023 - Present

University of California, Los Angeles

Marginalized Identities in Physics & Astronomy (MIPA)

Undergraduate Mentee & Mentor

September 2023 - Present

University of California, Los Angeles

AFFILIATIONS

American Physical Society (APS) Member

Society for the Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS) Member

Out in Science, Technology, Engineering & Mathematics (oSTEM) Member

Society of Physics Students (SPS) Member

UCLA Academic Advancement Program (AAP) Member

FIRST Tech Challenge (FTC) Alum

SKILLS

Technical

- Programming: Python, C++
- Other: $\text{\LaTeX}$, Git, Bash

RELEVANT COURSEWORK

Quantum Mechanics, Analytic Mechanics, Real Analysis (honors), Abstract Algebra: Rings, Ordinary Differential Equations, Linear Algebra (proof-based), Multivariable Calculus, Statistical Mechanics, Galactic Astrophysics, Fundamentals of Astrophysics, Optics, Electricity and Magnetism, Classical Mechanics, Introduction to Computer Science (C++).